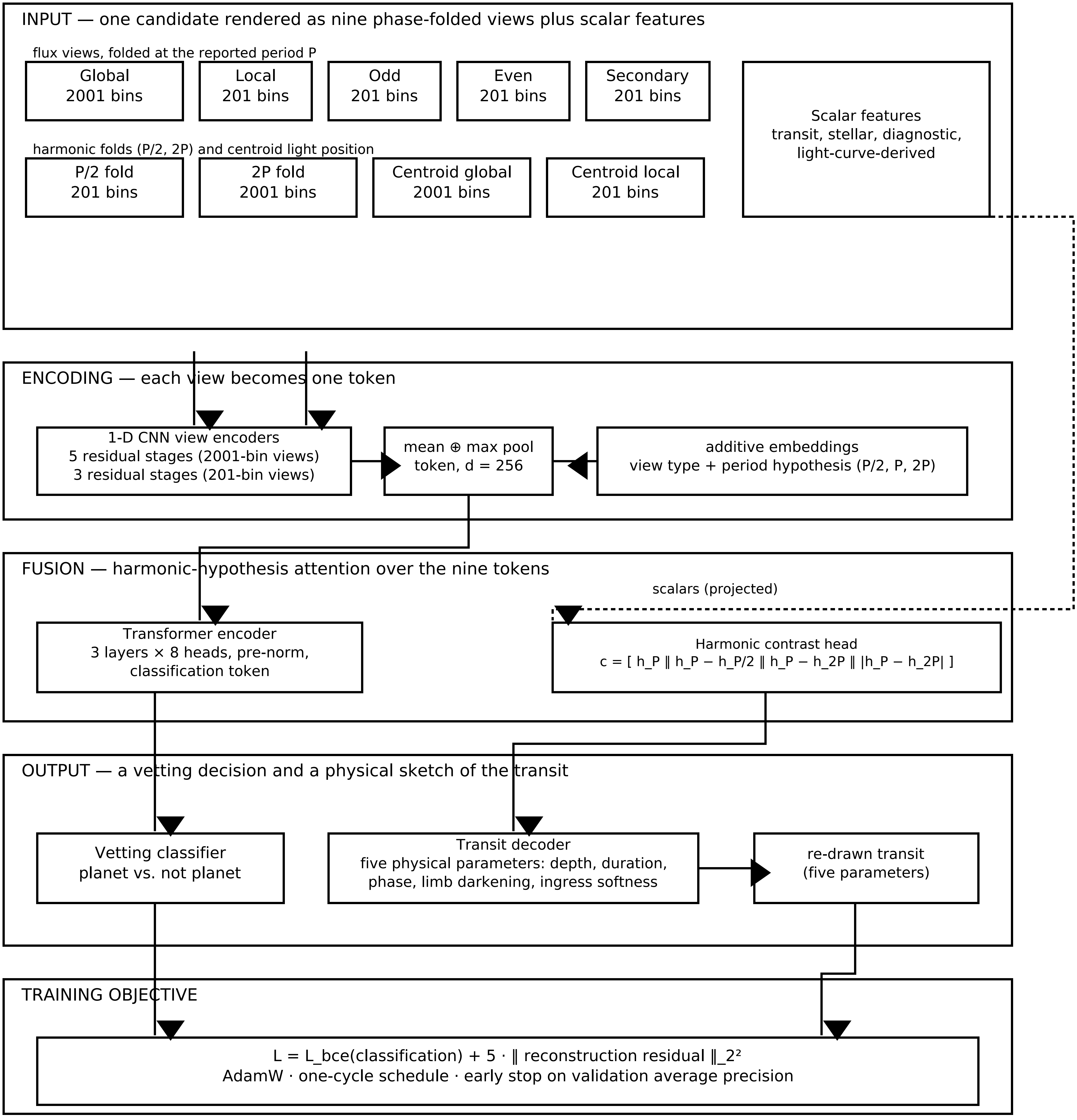
INPUT — one candidate rendered as nine phase-folded views plus scalar features
flux views, folded at the reported period P
Global
2001 bins
Local
201 bins
Odd
201 bins
Even
201 bins
Secondary
201 bins
Scalar features
transit, stellar, diagnostic,
light-curve-derived
harmonic folds (P/2, 2P) and centroid light position
P/2 fold
201 bins
2P fold
2001 bins
Centroid global
2001 bins
Centroid local
201 bins
ENCODING — each view becomes one token
1-D CNN view encoders
5 residual stages (2001-bin views)
3 residual stages (201-bin views)
mean ⊕ max pool
token, d = 256
additive embeddings
view type + period hypothesis (P/2, P, 2P)
FUSION — harmonic-hypothesis attention over the nine tokens
scalars (projected)
Transformer encoder
3 layers × 8 heads, pre-norm,
classification token
Harmonic contrast head
c = [h_P ‖ h_P − h_P/2 ‖ h_P − h_2P ‖ |h_P − h_2P|]
OUTPUT — a vetting decision and a physical sketch of the transit
Vetting classifier
planet vs. not planet
Transit decoder
five physical parameters: depth, duration,
phase, limb darkening, ingress softness
re-drawn transit
(five parameters)
TRAINING OBJECTIVE
L = L_bce(classification) + 5 · ‖ reconstruction residual ‖_2²
AdamW · one-cycle schedule · early stop on validation average precision